

BUSINESS REQUIREMENTS DOCUMENT

ChemTrack — Chemical Inventory & Invoice Automation System

Chemical Supply Business, Nagpur, India | Prepared by Vibhor Dharmadhikari, Business Analyst & Developer | Version 1.0 | 2024

DOCUMENT CONTROL

Document Title	ChemTrack — Business Requirements Document
Version	1.0 — Initial Release
Status	Final — Client Accepted
Prepared By	Vibhor Dharmadhikari — Business Analyst & Developer
Client	Chemical Supply Business — Nagpur, India (name withheld for confidentiality)
Engagement Type	Freelance Project
Date	2025
Platform	Microsoft Excel (Macro-Enabled Workbook .xlsm)

1. EXECUTIVE SUMMARY

A chemical supply business based in Nagpur, India — operating with a team of five including the owner — approached me to automate their order processing and invoice generation workflow. The business sourced chemicals from suppliers and fulfilled orders for a range of business clients. Their entire order-to-invoice process was manual: incoming orders arrived in multiple unstructured formats (Excel files, photographs of handwritten lists, and plain-text emails), and staff were required to manually cross-reference each order against a master inventory file to look up prices before typing a final invoice from scratch.

This process was consuming several hours of staff time daily, introducing pricing errors — particularly around products available in multiple variants (e.g. the same chemical in powder and liquid form, each with multiple size options) — and causing delays that occasionally led clients to source from competitors.

This document defines the business requirements for ChemTrack — a Microsoft Excel-based macro-enabled inventory and invoice automation system. ChemTrack consolidates the master product inventory and the invoice generation process into a single workbook, automating price lookup, flagging multi-variant products for manual review, and producing print-ready invoices from standardised order input.

2. BUSINESS CONTEXT & PROBLEM STATEMENT

2.1 Current State — As-Is Process

Prior to ChemTrack, the client's order-to-invoice workflow operated as follows:

1. A client places an order via one of three channels: an Excel file attached to an email, a photograph of a handwritten order list sent via WhatsApp, or a plain-text email listing product names and quantities
2. A staff member manually transcribes the order into a new Excel invoice sheet — retyping product names, quantities, and units from the order source
3. The staff member opens the master inventory file (a separate Excel workbook maintained by the owner) and manually searches for each product to find the listed unit price
4. The unit price is manually typed into the invoice sheet alongside the corresponding product line
5. The staff member calculates line totals and grand total manually or using basic Excel formulas
6. The completed invoice is printed and handed to the delivery driver or emailed to the client

Note: Tax is not applied at invoice stage — it is calculated separately at the point of final purchase transaction.

2.2 Business Problems Identified

- **Time Consumption:** Manual transcription and price lookup consumed an estimated 3–4 hours of staff time per day, scaling with order volume
- **Pricing Errors — Multi-Variant Products:** Several chemicals are available in multiple forms (powder vs liquid) and multiple sizes. Manual lookup frequently resulted in the wrong variant's price being applied — generating incorrect invoices that required time-consuming correction
- **Non-Standardised Order Input:** Incoming orders arrived in three different formats with no consistent structure — making transcription slow and error-prone
- **Client Loss Risk:** Invoice delays caused by manual processing led some clients to place orders with competitors, representing direct revenue loss
- **No Audit Trail:** No record was maintained of which staff member processed which order or what price was applied at the time of invoicing

2.3 Business Objectives

7. Automate price lookup from the master inventory — eliminating manual cross-referencing between separate files
8. Flag multi-variant products requiring manual price verification — preventing incorrect pricing while maintaining staff oversight for ambiguous entries
9. Standardise the order entry interface — providing a consistent input format regardless of how the client order arrived
10. Generate a print-ready invoice automatically from entered order data
11. Reduce order-to-invoice processing time by at least 70%

3. PROJECT SCOPE

3.1 In Scope

- Master inventory sheet — consolidated product list with product name, variant details, unit, and unit price
- Order entry sheet — standardised interface for staff to enter client order details (product name, quantity, unit)
- Automated price lookup — macro maps entered product names against the master inventory and populates unit prices automatically
- Multi-variant product flagging — when a product name matches multiple inventory entries (e.g. different sizes or forms), the price field is left blank and flagged for manual staff review
- Invoice generation sheet — auto-populated from order entry data, showing product name, quantity, unit, unit price, and line total
- Grand total calculation — automatic sum of all line totals on the invoice
- Print-ready invoice formatting — invoice sheet formatted for direct printing without further adjustment
- Single Excel workbook (.xlsm) — all functionality contained within one macro-enabled file

3.2 Out of Scope

- Tax calculation — applied externally at point of final purchase; not included in the invoice
- Client management — no client database, contact records, or order history tracking
- Stock level tracking — ChemTrack manages pricing only; physical stock levels are managed separately
- Multi-user access control or password protection — all five users access the same shared workbook
- Integration with any external system, email client, or accounting software
- Mobile or web-based interface — Excel desktop only
- Automated import of client order files — order data is manually entered by staff regardless of incoming format

4. STAKEHOLDERS

Role	Responsibilities & System Interaction	Count
Business Owner	Primary client and project sponsor. Maintains master inventory pricing. Final approver of invoice output. Participated in all requirements sessions and prototype reviews.	1
Staff — Order Processing	Day-to-day users of ChemTrack. Transcribe incoming client orders into the order entry sheet, review flagged multi-variant items, and print final invoices.	4
Business Analyst / Developer	Vibhor Dharmadhikari — conducted requirements elicitation sessions, analysed the master inventory, designed the workbook architecture, developed macros, and facilitated prototype reviews with the owner.	1
End Clients (Indirect)	Businesses placing chemical orders. Not direct system users but recipients of the printed invoices generated by ChemTrack. Invoice accuracy and speed directly impacts client satisfaction.	Multiple

5. REQUIREMENTS ELICITATION SUMMARY

Requirements for ChemTrack were gathered through a series of in-person sessions with the business owner, supplemented by direct analysis of the existing master inventory file and a sample client order provided on the first day of engagement.

5.1 Elicitation Sessions

Session	Method	Key Outputs
Session 1	Stakeholder interview + document analysis	Current process walkthrough documented; master inventory file and sample client order received and analysed; initial scope defined
Session 2	Prototype review	First working prototype demonstrated to owner; multi-variant product issue identified — same chemical available in powder/liquid and multiple sizes causing incorrect price matching; flagging requirement captured

Session 3	UAT walkthrough + sign-off	Owner and staff tested the system using live order data; flagging behaviour confirmed as accepted solution; print layout approved; system accepted
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5.2 Key Discovery — Multi-Variant Product Issue

During the first prototype review, a critical data quality issue was identified. The master inventory supplied by the owner — originally received from their chemical supplier — contained multiple entries for the same product name where variants differed by form (powder vs liquid) and size (e.g. 500g, 1kg, 5kg). When a client order listed only a product name without specifying variant details, the automated price lookup could not reliably determine which entry to match — risking incorrect pricing.

This discovery drove the design of the multi-variant flagging business rule (BR-002), which was agreed with the owner as the appropriate solution — preserving automation for unambiguous products while routing ambiguous entries to staff for manual review.

6. FUNCTIONAL REQUIREMENTS

Requirements are classified by priority: High (must have for delivery), Medium (important, included in v1.0), Low (future consideration).

ID	Requirement Description	Priority	Source
FR-001	The system shall maintain a Master Inventory sheet listing all chemical products with the following fields: Product Name, Variant Description (form and size), Unit of Measure, and Unit Price	High	Owner
FR-002	The Master Inventory shall support multiple entries for the same product name where variants differ — each variant listed as a separate row with its own Unit Price	High	Prototype Review
FR-003	The Master Inventory shall be editable by the owner to update prices, add new products, or deactivate discontinued variants	High	Owner
FR-004	The system shall provide a dedicated Order Entry sheet with the following input fields per line: Product Name, Quantity, and Unit of Measure	High	Owner
FR-005	The Order Entry sheet shall support a minimum of 50 line items per order to accommodate large client orders	High	Owner
FR-006	The system shall include a Client Name field and Order Date field at the top of the Order Entry sheet	Medium	Owner
FR-007	Upon entry of a Product Name in the Order Entry sheet, the system macro shall automatically search the Master Inventory for a matching product and populate the Unit Price field	High	Owner
FR-008	The price lookup shall be case-insensitive and shall trim leading/trailing spaces to account for staff entry variations	Medium	BA Analysis
FR-009	If a product name matches exactly ONE entry in the Master Inventory, the Unit Price shall be populated automatically without staff intervention	High	Owner
FR-010	If a product name matches MULTIPLE entries in the Master Inventory (multi-variant product), the Unit Price field shall be left blank and a visible flag indicator shall appear on that row alerting staff to manually verify and enter the correct price	High	Prototype Review
FR-011	If a product name matches NO entries in the Master Inventory, the Unit Price field shall display 'Product Not Found' and the row shall be flagged for staff review	High	BA Analysis

FR-012	The system shall auto-populate an Invoice sheet from the Order Entry data, displaying: Client Name, Order Date, Product Name, Variant (where applicable), Quantity, Unit of Measure, Unit Price, and Line Total	High	Owner
FR-013	Line Total shall be calculated automatically as Quantity multiplied by Unit Price for all rows where Unit Price has been populated	High	Owner
FR-014	The Invoice sheet shall display a Grand Total — the sum of all Line Totals — at the bottom of the invoice	High	Owner
FR-015	Tax shall NOT be calculated or displayed on the invoice — a note stating 'Tax applicable at point of purchase' shall appear below the Grand Total	High	Owner
FR-016	Rows with flagged items (blank or unresolved Unit Price) shall be visually distinguishable on the Invoice sheet — e.g. highlighted in a different colour — to alert staff before printing	High	Prototype Review
FR-017	The Invoice sheet shall be formatted for A4 printing with appropriate margins, font sizes, and page breaks such that a standard invoice prints cleanly without manual formatting adjustment	High	Owner
FR-018	The workbook shall include a Print Invoice button that triggers direct printing of the Invoice sheet	Medium	Owner
FR-019	The workbook shall include a Clear Order button that resets the Order Entry sheet for a new order without affecting the Master Inventory data	High	Staff Feedback
FR-020	The macro shall run automatically when the staff member moves to the next row after entering a product name — no manual macro trigger required	Medium	Staff Feedback

7. NON-FUNCTIONAL REQUIREMENTS

7.1 Performance

- The price lookup macro shall complete execution within 2 seconds for orders of up to 50 line items on standard office hardware
- The system shall function without performance degradation on Microsoft Excel 2016 or later

7.2 Usability

- The system shall be operable by staff with basic Excel proficiency — no VBA knowledge or technical training required
- All flagged items shall be identifiable at a glance without requiring staff to open any additional files or run any manual checks
- Instructions for common tasks (entering an order, handling flagged items, printing an invoice, clearing for a new order) shall be included on a dedicated Help sheet within the workbook

7.3 Reliability

- The macro shall not alter or overwrite any data in the Master Inventory sheet under any circumstances
- The system shall handle blank rows in the Order Entry sheet gracefully — skipping them without generating errors
- The workbook shall be saved in .xlsm format to preserve macro functionality across sessions

7.4 Maintainability

- The Master Inventory sheet shall be structured so the owner can add, edit, or remove product entries without requiring developer assistance
- All macro code shall be commented in plain language so future modifications can be made by a developer without prior project knowledge

8. KEY BUSINESS RULES

ID	Business Rule	Detail
BR-001	No Tax on Invoice	Tax is explicitly excluded from the invoice. A note stating 'Tax applicable at point of purchase' must appear on every printed invoice below the Grand Total.
BR-002	Multi-Variant Flagging	If a product name matches more than one Master Inventory entry, the Unit Price field is left blank and flagged. The staff member must manually look up and enter the correct price. The system must never auto-select a price where ambiguity exists.
BR-003	Master Inventory Is Read-Only During Processing	The price lookup macro must read from the Master Inventory but must never write to, modify, or reformat it. All macro outputs are written to the Order Entry and Invoice sheets only.
BR-004	Unresolved Flags Block Clean Print	Any invoice line with an unresolved flag (blank Unit Price) shall remain visually highlighted on the Invoice sheet. Staff must resolve all flags before printing to ensure no zero-priced lines appear on client invoices.
BR-005	Clear Order Resets Entry Only	The Clear Order function resets the Order Entry sheet and Invoice sheet only. The Master Inventory sheet is never affected by any user action other than direct manual editing by the owner.

9. ASSUMPTIONS & DEPENDENCIES

9.1 Assumptions

- The Master Inventory file provided by the owner is complete and accurate at the time of system delivery — ChemTrack's accuracy is dependent on the quality of inventory data
- All five users have access to a Windows PC running Microsoft Excel 2016 or later with macros enabled
- Staff entering orders have sufficient product knowledge to resolve multi-variant flags by manually referencing the Master Inventory
- The owner will maintain the Master Inventory sheet on an ongoing basis as prices change or new products are added
- Client orders will continue to arrive in non-standardised formats — ChemTrack standardises the internal processing step but does not change how clients submit orders

9.2 Dependencies

- Owner to provide complete and formatted Master Inventory file before development of the price lookup macro
- Owner to provide at least three representative sample client orders (one Excel, one photo/transcribed, one email-based) for use in prototype testing
- Staff availability for UAT session — minimum 30 minutes with two staff members entering live order data

10. RISKS & MITIGATIONS

Risk	Impact	Mitigation
Master Inventory data quality — product names inconsistently formatted across entries	High — price lookup failures or incorrect matches	BA to analyse inventory file before development and flag inconsistencies to owner for correction. Lookup macro to include case-insensitive and whitespace-trim logic (FR-008).

Multi-variant products not fully identified upfront — new variants discovered post-delivery	Medium — flagging rule works correctly but owner may expect auto-resolution for variants not yet mapped	BR-002 clearly communicated to owner at sign-off. Owner instructed to maintain variant entries in Master Inventory as new product sizes are added.
Staff disable macros on opening the workbook — common Excel security prompt	High — all automation non-functional without macros enabled	Help sheet to include clear instructions for enabling macros. Owner to set file as trusted location on all office PCs before go-live.
Scope creep — owner requests stock level tracking or client order history mid-engagement	Medium — timeline and complexity risk for a fixed-fee freelance project	Scope documented clearly in Section 3.2 as out of scope. Any additions to be scoped and priced separately.

11. ACCEPTANCE CRITERIA

The following criteria were used to confirm client acceptance of ChemTrack at the UAT session:

- Price lookup correctly auto-populates Unit Price for all single-match products across a test order of 20+ line items
- Multi-variant products (minimum 5 tested) correctly trigger the flag indicator and leave Unit Price blank — no incorrect auto-population occurs
- Products not found in the Master Inventory correctly display 'Product Not Found' and are flagged
- Invoice sheet auto-populates correctly from Order Entry data with accurate Line Totals and Grand Total
- Tax note appears on every invoice below the Grand Total
- Flagged rows remain visually highlighted on the Invoice sheet until manually resolved
- Print Invoice button produces a clean A4 print without manual formatting
- Clear Order button resets Order Entry and Invoice sheets without affecting Master Inventory
- Owner confirms Master Inventory remains unaltered after 10 complete order processing cycles
- Business owner formal verbal sign-off received at UAT session completion

12. DELIVERED OUTCOMES

ChemTrack was delivered, tested, and accepted by the client following the UAT session. The following outcomes were confirmed post-delivery:

- Order-to-invoice processing time reduced significantly — staff estimated a reduction from 3–4 hours per day to under 1 hour for equivalent order volumes
- Pricing errors on multi-variant products eliminated through the flagging mechanism, staff now have a clear, consistent process for handling ambiguous entries
- All five users (owner and four staff) adopted the system without requiring technical training beyond a brief walkthrough
- The owner successfully maintained the Master Inventory independently following delivery — adding new product entries and updating prices without developer assistance

13. DOCUMENT SIGN-OFF

Name	Role	Signature	Date
Vibhor Dharmadhikari	Business Analyst & Developer		2025
Client	Project Sponsor	Verbal sign-off at UAT	2025